

Solutions of TPBVP Using Indirect method

Shooting method (Quasi-linearization method)

$$J = h(x(t_f), t_f) + \int_0^{t_f} L(x, u, t) dt$$

$$\dot{x} = f(x, u, t)$$

$$x(0) = \text{given}, \quad x(t_f) = \text{given}$$

$$\dot{\lambda} = -\frac{\partial H}{\partial x} = g(x, u, \lambda, t) \quad \lambda(t_0) \text{ is guessed}$$

$$\frac{\partial H}{\partial u} = 0 \quad \rightarrow u = u(x, \lambda)$$

Substitute $u = u(x, \lambda)$ into $\dot{x} = f(x, u, t)$ and $\dot{\lambda} = g(x, u, \lambda, t)$, then we have

$$\dot{x} = f(x, \lambda, t) \tag{1}$$

$$\dot{\lambda} = g(x, \lambda, t) \tag{2}$$

$$\text{Let } X = \begin{Bmatrix} x \\ \lambda \end{Bmatrix}$$

$$\dot{X} = \begin{Bmatrix} \dot{x} \\ \dot{\lambda} \end{Bmatrix} = \begin{Bmatrix} f \\ g \end{Bmatrix} \equiv F(X, t) \tag{3}$$

The set of differential equations eq. (3) are linearized about a nominal solution $X^k(t)$ for k^{th} trial solution (i.e., the right-hand side of eq (3) is expanded in a Taylor's series and only first-order terms are retained.). The linearized equations are given by

$$\begin{aligned} \dot{X}^k(t) + \Delta \dot{X}^k(t) &= F(X^k, t) + \left. \frac{\partial F(X, t)}{\partial X} \right|_{X^k} \Delta X^k \\ \Delta \dot{X}^k(t) &= \left. \frac{\partial F(X, t)}{\partial X} \right|_{X^k} \Delta X^k + \underbrace{\left[F(X^k, t) - \dot{X}^k(t) \right]}_{\equiv d(t)} \end{aligned} \tag{4}$$

$$\text{Let } Y = \begin{Bmatrix} \Delta x^k \\ \Delta \lambda^k \end{Bmatrix} = \{\Delta X^k\}$$

$$\frac{\partial F}{\partial X} \Big|_{X^k} \equiv A(t) \quad \text{then}$$

$$\dot{Y}(t) = A(t)Y(t) + d(t), \quad d(t) \text{ is going to zero.} \quad (5)$$

Approximate

$$\dot{Y}(t) = A(t)Y(t) \quad (6)$$

This is a linear equation then

$$Y(t) = \Phi(t, t_0)Y(t_0) \quad (7)$$

$$\text{Integrate the eq (5) forward in time using } Y(t_0) = \begin{bmatrix} \Delta x_1(t_0) \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Then, what we get is the first column of $\Phi(t_f, t_0)$ as following.

$$Y(t_f) = \Phi(t_f, t_0)Y(t_0) = \begin{bmatrix} \Phi_{11} & \Phi_{12} & \cdots & \Phi_{12n} \\ \Phi_{21} & \ddots & & \vdots \\ \vdots & & \ddots & \Phi_{2n-1, 2n} \\ \Phi_{2n1} & \cdots & \Phi_{2n, 2n-1} & \Phi_{2n, 2n} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} \Phi_{11} \\ \Phi_{21} \\ \vdots \\ \Phi_{2n1} \end{bmatrix}$$

Similarly to get the second column of $\Phi(t_f, t_0)$,

$$Y(t_0) = \begin{bmatrix} 0 \\ \Delta x_2(t_0) \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} \Phi_{11} & \Phi_{12} & \cdots & \Phi_{12n} \\ \Phi_{21} & \ddots & & \vdots \\ \vdots & & \ddots & \Phi_{2n-1, 2n} \\ \Phi_{2n1} & \cdots & \Phi_{2n, 2n-1} & \Phi_{2n, 2n} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} \Phi_{12} \\ \Phi_{22} \\ \vdots \\ \Phi_{2n2} \end{bmatrix}$$

Iteration

Finally we have $Y(t_f) = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} = \Phi(t_f, t_0)$ $2n$ by $2n$ matrix

$$\begin{bmatrix} \Delta x(t_f) \\ \Delta \lambda(t_f) \end{bmatrix}_{2n \times 1} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix}_{2n \times 2n} \begin{bmatrix} \Delta x(t_0) \\ \Delta \lambda(t_0) \end{bmatrix}_{2n \times 1}$$

$$\Delta x(t_f) = \phi_{11} \Delta x(t_0) + \phi_{12} \Delta \lambda(t_0)$$

$$\rightarrow \Delta \lambda(t_0) = \phi_{12}^{-1} [\Delta x(t_f) - \phi_{11} \Delta x(t_0)]$$

$$\therefore \lambda^{k+1}(t_0) = \lambda^k(t_0) + \Delta \lambda^k(t_0)$$

If we integrate the original state and costate equations forward in time, with $x(t_0)$ given and $\lambda(0) = \lambda^{k+1}(t_0)$, we will get a new solution to linearize about x, λ . Then we solve another linear TPBVP with convergence ($d(t) = 0$)

Algorithm

- 1) Guess the initial costate condition

$$\lambda^k(t_0)$$

- 2) Using $\frac{\partial H}{\partial u} = 0$ equation, control $u(t)$ is calculated in terms of x, λ and u is eliminated in $\dot{x}, \dot{\lambda}$ equations.

$$\begin{aligned} \text{Set } \dot{x}^k &= f(x^k, \lambda^k, t) \\ \dot{\lambda}^k &= g(x^k, \lambda^k, t) = -\frac{\partial H}{\partial x} \end{aligned}$$

- 3) Integrate $\dot{x}^k = f, \dot{\lambda}^k = g$ forward in time using $x(t_0)$ and guessed $\lambda^k(0)$.
- 4) Check $\Delta x(t_f) < \gamma$ (tolerance)

If $\Delta x(t_f) \equiv x^k(t_f) - x(t_f)$, $x^k(t_f)$: calculated, $x(t_f)$: given

If $\Delta x(t_f) < \gamma$, stop, else go to 5

5) Linearize about $x^k(t)$, $\lambda^k(t)$

$$\Delta \dot{x}^k = f'(x^k, \lambda^k, \Delta x^k, \Delta \lambda^k, t)$$

$$\Delta \dot{\lambda}^k = g'(x^k, \lambda^k, \Delta x^k, \Delta \lambda^k, t)$$

6) Obtain the state transition matrix by perturbing $\begin{bmatrix} \Delta x^k(t_0) \\ \Delta \lambda^k(t_0) \end{bmatrix}$

7) Get $\Delta \lambda^k(t_0) = \phi_{12}^{-1} [\Delta x(t_f) - \phi_{11} \Delta x(t_0)]$

$$\lambda^{k+1}(t_0) = \lambda^k(t_0) + \Delta \lambda^k(t_0)$$

Goto step 2)

Shooting method for optimization of Dynamic System with free parameters (constant)

$$\dot{x} = f(x, u, p, t)$$

$x(0)$ and $x(t_f)$ are given

$$J = h(x(t_f), t_f) + \int_0^{t_f} g(x, u, p, t) dt$$

$$\begin{aligned} \delta J = & \left[\frac{\partial h}{\partial x(t_f)} - \Lambda(t_f) \right] \delta x_f \\ & + \int_0^{t_f} \left[\frac{\partial H}{\partial u} \delta u + \left(\frac{\partial H}{\partial x} + \dot{\Lambda} \right) \delta x + \left(\frac{\partial H}{\partial \Lambda} - \dot{x} \right) \delta \Lambda + \frac{\partial H}{\partial p} \delta p \right] dt \end{aligned}$$

$$\delta J = \int_0^{t_f} \frac{\partial H}{\partial p} \delta p dt = 0$$

$$\rightarrow \delta p \int_0^{t_f} \frac{\partial H}{\partial p} dt = 0 \quad \delta p : \text{constant at initial time}$$

$$\rightarrow \int_0^{t_f} \frac{\partial H}{\partial p} dt = 0$$

$$\text{Let } \dot{z} = -\frac{\partial H}{\partial p}$$

$$\int_0^{t_f} \frac{\partial H}{\partial p} dt \rightarrow -[z(t_f) - z(0)] = 0$$

After using the optimality condition $\frac{\partial H}{\partial u} = 0$, we eliminate u in terms of Λ, x

$$\dot{x} = f(x, u, p, t) \quad , \quad x \in \mathbf{R}^n$$

$$\dot{\Lambda} = -\frac{\partial H}{\partial x} \quad , \quad \Lambda \in \mathbf{R}^n$$

$$\dot{z} = -\frac{\partial H}{\partial p} \quad , \quad p \in \mathbf{R}^m$$

$$z(0) = 0, \quad z(t_f) = 0 \quad , \quad z \in \mathbf{R}^m$$

We are looking for values of $\Lambda(0), p$

$$\dot{\Lambda} = g'(x, \Lambda, t) \quad \leftarrow \partial H / \partial u = 0, \quad u = u(x, \Lambda)$$

$$X_1 = \begin{Bmatrix} x \\ z \end{Bmatrix}, \quad X_2 = \{\Lambda\}$$

$$X = \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix} = \begin{Bmatrix} x \\ z \\ \Lambda \end{Bmatrix}$$

$$\dot{X} = \begin{Bmatrix} \dot{x} \\ \dot{z} \\ \dot{\Lambda} \end{Bmatrix} \equiv F(X, p, t)$$

1) Guess the initial value of $\Lambda(0)$ and p

2) Linearize the state-costate equation about the nominal solution generated from forward integration of the nonlinear equation.

$$\Delta \dot{X} = \frac{\partial F}{\partial X} \Delta X + \frac{\partial F}{\partial p} \Delta p$$

$$\Delta X(t_f) = \Phi \Delta X(0) + \psi \Delta p$$

where Φ is a $(2n+m) \times (2n+m)$ matrix

ψ is a $(2n+m) \times m$ matrix

$$\Phi_{ij} = \frac{\partial \Delta X_i(t_f)}{\partial \Delta X_j(0)}, \quad \psi_{ij} = \frac{\partial \Delta X_i(t_f)}{\partial \Delta p_j}$$

Method A: Finite Difference Method

Change the first state variable slightly from its nominal value.

Hold all other quantities at their nominal values

$$x_1(0) = x_1^N(0) + \varepsilon$$

$$x_i(0) = x_i^N(0), \quad i \neq 1$$

Integrate the $\dot{x}, \dot{\Lambda}, \dot{z}$ equation forward in time.

Note the value of $X(t_f)$

$$\Phi_{ij} = \frac{X(t_f) - X^N(t_f)}{\varepsilon}$$

To get ψ_{ij} , do the same thing with p

Method B:

$$\Delta X(0) = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \Delta p = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Integrate $\Delta \dot{X}$ to get $\Delta X(t_f)$

$$\Rightarrow \Delta x(t_f) = \Phi_{i1}$$

Repeating yields Φ_{ij} matrix.

Similarly, follow the same step with Δp to get ψ_{ij} matrix.

Method B gives more accurate results than F.D.M, but requires more work (analytical partial derivative).

3) After finding Φ and ψ , we are ready to solve for the missing $\Delta X(0)$ and Δp .

$$\begin{bmatrix} \Delta X_1(t_f) \\ \Delta \Lambda(t_f) \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} \Delta X_1(0) \\ \Delta \Lambda(0) \end{bmatrix} + \begin{bmatrix} \psi_1 \\ \psi_2 \end{bmatrix} \Delta p$$

$$\begin{aligned} \Delta X_1(t_f) &= \phi_{11} \Delta X_1(0) + \phi_{12} \Delta \Lambda(0) + \psi_1 \Delta p \\ &= [\phi_{12} \quad \psi_1] \begin{bmatrix} \Delta \Lambda(0) \\ \Delta p \end{bmatrix} \end{aligned}$$

$$\therefore \begin{bmatrix} \Delta \Lambda(0) \\ \Delta p \end{bmatrix} = [\phi_{12} \quad \psi_1]^{-1} \Delta X_1(t_f)$$

Example: Solve the following Optimal control Problem numerically using a shooting method.

$$\text{Minimize } J = \int_0^{t_f} (1 + u^2) dt$$

$$\text{Subject to } \dot{x} = -x + x^3 + u$$

$$x(0) = 1, \quad x(t_f) = 0, \quad t_f \text{ is free}$$

Let's convert to fixed time problem.

$$\text{Define } t = p^2 \tau$$

$$dt = p^2 d\tau$$

where τ is non dimensional time $\tau \in [0,1]$

$$\frac{d}{dt} = \frac{1}{p^2} \frac{d}{d\tau}$$

$$A' \equiv \frac{d}{d\tau} A$$

$$\text{Then, } \dot{x} = -x + x^3 + u \Leftrightarrow x' = (-x + x^3 + u)p^2$$

Let free parameter p be independent variable

$$\text{then, } p' = 0$$

$$\text{and } J = \int_0^1 (1 + u^2) p^2 d\tau$$

The Hamiltonian is

$$H = (1 + u^2)p^2 + \lambda_1(-x + x^3 + u)p^2 + \lambda_2 0$$

$$H(t_f) = \frac{\partial h}{\partial t_f} = 0$$

$$\lambda_1' = -\frac{\partial H}{\partial x} = -\lambda_1 p^2(-1 + 3x^2) = \lambda_1 p^2(1 - 3x^2)$$

$$\lambda_2' = -\frac{\partial H}{\partial p} = -2p(1 + u^2) - 2p\lambda_1(-x + x^3 + u)$$

$$0 = \frac{\partial H}{\partial u} = p^2(2u + \lambda_1) \quad , p \neq 0$$

$$\rightarrow u = -\frac{\lambda_1}{2}$$

$$\begin{array}{l} \text{State equation} \quad x' = (-x + x^3 - \frac{1}{2}\lambda_1)p^2 \\ \quad \quad \quad p' = 0 \end{array}$$

$$\begin{array}{l} \text{Costate equation} \quad \lambda_1' = \lambda_1 p^2(1 - 3x^2) \\ \quad \quad \quad \lambda_2' = -2p(1 + \frac{1}{4}\lambda_1^2) - 2p\lambda_1(-x + x^3 - \frac{1}{2}\lambda_1) \end{array}$$

$$p; \text{ parameter } \xrightarrow{\delta J=0} \int_0^1 \frac{\partial H}{\partial p} d\tau = 0$$

$$z(0) = z(1) = 0$$

$$z' = -\frac{\partial H}{\partial p}$$

Let $\lambda_1 = \lambda$, $\lambda_2 = z$

Then, we obtain

$$x' = \left(-x + x^3 - \frac{1}{2}\lambda\right)p^2 \quad (1-1)$$

$$z' = -2p\left(1 + \frac{1}{4}\lambda^2\right) - 2p\lambda\left(-x + x^3 - \frac{1}{2}\lambda\right) \quad (1-2)$$

$$\lambda' = \lambda p^2(1 - 3x^2) \quad (1-3)$$

$$\begin{aligned} x(0) &= 1, & x(1) &= 0 \\ z(0) &= 0, & z(1) &= 0 \end{aligned}$$

We are looking for values of $\lambda(0)$ and p

- 1) Guess the values of $\lambda(0)$ and p
- 2) Linearize the state-costate equations (1-1),(1-2),(1-3) about the nominal solution generated from forward integration of the nonlinear equations.

$$X \equiv \begin{bmatrix} x \\ z \\ \lambda \end{bmatrix}, \quad F \equiv \begin{bmatrix} F_1 \\ F_2 \\ F_3 \end{bmatrix}$$

$$X' = F(X, p)$$

$$\Delta X' = \frac{\partial F}{\partial X} \Delta X + \frac{\partial F}{\partial p} \Delta p$$

$$\frac{\partial F}{\partial X} = \begin{bmatrix} \frac{\partial F_1}{\partial x} & \frac{\partial F_1}{\partial z} & \frac{\partial F_1}{\partial \lambda} \\ \frac{\partial F_2}{\partial x} & \frac{\partial F_2}{\partial z} & \frac{\partial F_2}{\partial \lambda} \\ \frac{\partial F_3}{\partial x} & \frac{\partial F_3}{\partial z} & \frac{\partial F_3}{\partial \lambda} \end{bmatrix}, \quad \frac{\partial F}{\partial p} = \begin{bmatrix} \frac{\partial F_1}{\partial p} \\ \frac{\partial F_2}{\partial p} \\ \frac{\partial F_3}{\partial p} \end{bmatrix}$$

$$\begin{aligned} \begin{bmatrix} \Delta x' \\ \Delta z' \\ \Delta \lambda' \end{bmatrix} &= \begin{bmatrix} p^2(-1 + 3x^2) & 0 & -\frac{1}{2}p^2 \\ 2p\lambda(1 - 3x^2) & 0 & -2px + 2px^3 + p\lambda \\ -6\lambda p^2 x & 0 & p^2(1 - 3x^2) \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta z \\ \Delta \lambda \end{bmatrix} \\ &+ \begin{bmatrix} 2p(-x + x^3 - \frac{1}{2}\lambda) \\ -2 + \frac{1}{2}\lambda^2 + 2\lambda x - 2\lambda x^3 \\ 2p\lambda(1 - 3x^2) \end{bmatrix} \Delta p \end{aligned} \quad (2)$$

$$\Delta X(1) = \Phi \Delta X(0) + \Psi \Delta p \quad (3)$$

Let $\Delta X_i(0) = \varepsilon$ ($i=1$) and all other $\Delta X_j(0) = 0$ ($i \neq j$) and $\Delta p = 0$

Integrate equation (2) then, by equation (3)

$$\Delta X(1) = \Phi(*, i) \varepsilon$$

Doing this for all known $\Delta X_i(0)$, we obtain

$$\Delta X(1) = [\Phi(*, 1), \Phi(*, 2), \Phi(*, 3)] \Delta X(0)$$

Similarly follow the same step with Δp also, then we get Ψ

$$\Delta X(1) = \Psi \Delta p$$

After finding Φ and Ψ , we are ready to solve for missing $\Delta X(0)$ and Δp .

$$\begin{bmatrix} \Delta x(1) \\ \Delta z(1) \\ \Delta \lambda(1) \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} & \phi_{13} \\ \phi_{21} & \phi_{22} & \phi_{23} \\ \phi_{31} & \phi_{32} & \phi_{33} \end{bmatrix} \begin{bmatrix} \Delta x(0) \\ \Delta z(0) \\ \Delta \lambda(0) \end{bmatrix} + \begin{bmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \end{bmatrix} \Delta p$$

$$\begin{bmatrix} \Delta x(1) \\ \Delta z(1) \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} \Delta x(0) \\ \Delta z(0) \end{bmatrix} + \begin{bmatrix} \phi_{13} \\ \phi_{23} \end{bmatrix} \Delta \lambda(0) + \begin{bmatrix} \psi_1 \\ \psi_2 \end{bmatrix} \Delta p$$

$$\therefore \begin{bmatrix} \Delta \lambda(0) \\ \Delta p \end{bmatrix} = \begin{bmatrix} \phi_{13} & \psi_1 \\ \phi_{23} & \psi_2 \end{bmatrix}^{-1} \begin{bmatrix} \Delta x(1) \\ \Delta z(1) \end{bmatrix}$$

$$\therefore \lambda^{new}(0) = \lambda^{old}(0) + \Delta \lambda(0)$$

$$p^{new} = p^{old} + \Delta p$$

Repeat above procedure until

$$\Delta x(1) = x_g(1) - x_c(1) = 0$$

$$\Delta z(1) = z_g(1) - z_c(1) = 0$$

The answer, $p=0.9822 \rightarrow t_f = p^2 = 0.9648$

The Nonlinear Programming Problem

Optimization problem involves constraints. If both the performance index and constraints are linear functions of the variables, the optimization problem is referred to as a linear programming problem. When nonlinear terms are present in the constraints or in performance index the problem is referred to as a nonlinear programming problem.

Consider the problem of extremizing the nonlinear performance index p

$$p = f(\vec{x}) \quad (1)$$

Where the real x_i of $\vec{x} = [x_1, x_2, \dots, x_n]^T$ are constrained by the following sets of equations;

$$g_i(\vec{x}) = C_i, \quad i = 1, 2, \dots, k < n \quad (2)$$

$$g_i(\vec{x}) \leq C_i, \quad i = k+1, k+2, \dots, m \quad (3)$$

All C_i are constraints, and all $g_i(\vec{x})$ are assumed to be of class C^1 and real values.

A) Unconstrained cases

$$\min f(\vec{x})$$

1) First-order Gradient method

$$\vec{x}^{new} = \vec{x}^{old} - \alpha \frac{\partial f}{\partial \vec{x}}$$

2) Second order Gradient Method

$$f(\bar{x}^k) \approx f(\bar{x}^*) + \underbrace{\mathbf{G}^k}_{\text{Jacobian}} \Delta \bar{x}^k + \frac{1}{2} \Delta \bar{x}^k \underbrace{\mathbf{H}^k}_{\text{Hessian}} \Delta \bar{x}^k$$

$\bar{x}^k$; current guess

Minimize $\Delta f \approx f(\bar{x}^k) - f(\bar{x}^*)$ with respect to $\Delta \bar{x}^k$

$$\rightarrow \frac{\partial \Delta f}{\partial \Delta \bar{x}} = 0$$

$$\mathbf{G}^k + \mathbf{H}^k \Delta \bar{x}^k = 0 \rightarrow \Delta \bar{x}^k = -[\mathbf{H}^k]^{-1} \mathbf{G}^k$$

$$\bar{x}^{k+1} = \bar{x}^k - \alpha [\mathbf{H}^k]^{-1} \mathbf{G}^k$$

Check decent

$$\Delta f(\bar{x}) \approx \frac{\partial f}{\partial \bar{x}} \Delta \bar{x}^k = -\alpha \mathbf{G}^k [\mathbf{H}^k]^{-1} \mathbf{G}^k < 0$$

If $\mathbf{H}^k$ is positive definite, it is expensive to evaluate $[\mathbf{H}^k]^{-1}$

3) Quasi-Newton Method

; Approximate the inverse of the Hessian matrix.

The search direction is still along the negative gradient

$$\Delta \bar{x}^k \equiv -\mathbf{A}^k \mathbf{G}^k$$

where $\mathbf{A}^k \approx [\mathbf{H}^k]^{-1}$ approximation to the inverse of the Hessian matrix.

$$\mathbf{A}^{k+1} = \mathbf{A}^k + \Delta \mathbf{A}^k, \quad \mathbf{A}^0 = \mathbf{I}$$

We want to update $\mathbf{A}^k$ such that

$$\Delta \bar{x}^k = \mathbf{A}^{k+1} \underbrace{[\mathbf{G}^{k-1} - \mathbf{G}^k]}_{=\Delta \mathbf{G}^k} \Rightarrow \underbrace{\mathbf{A}^{k+1}}_{\mathbf{A}^k + \Delta \mathbf{A}^k} = \left[\frac{\Delta \mathbf{G}^k}{\Delta \bar{x}^k} \right]^{-1} \text{ looks like inverse Hessian}$$

$$\begin{aligned}\Delta \bar{\mathbf{x}}^k &= \mathbf{A}^{k+1} \Delta \mathbf{G}^k = (\mathbf{A}^k + \Delta \mathbf{A}^k) \Delta \mathbf{G}^k \\ \Delta \mathbf{A}^k \Delta \mathbf{G}^k &= \Delta \bar{\mathbf{x}}^k - \mathbf{A}^k \Delta \mathbf{G}^k \\ &\rightarrow \Delta \mathbf{A}^k = \frac{\Delta \bar{\mathbf{x}}^k}{\Delta \mathbf{G}^k} - \frac{\mathbf{A}^k \Delta \mathbf{G}^k}{\Delta \mathbf{G}^k}\end{aligned}$$

The general solution to the above equation is

$$\Delta \mathbf{A}^k = \frac{\Delta \bar{\mathbf{x}}^k \bar{\mathbf{y}}^T}{\bar{\mathbf{y}}^T \Delta \mathbf{G}^k} - \frac{\mathbf{A}^k \Delta \mathbf{G}^k \bar{\mathbf{z}}^T}{\bar{\mathbf{z}}^T \Delta \mathbf{G}^k}$$

for arbitrary $\bar{\mathbf{y}}$ & $\bar{\mathbf{z}}$

DFP (Davidan, Flerher, Dowell) suggest

$$\bar{\mathbf{y}} = \Delta \bar{\mathbf{x}}^k, \quad \bar{\mathbf{z}} = \mathbf{A}^k \Delta \mathbf{G}^k$$

$$\therefore \mathbf{A}^{k+1} = \mathbf{A}^k + \Delta \mathbf{A}^k$$

$$\Delta \mathbf{A}^k = \frac{\Delta \bar{\mathbf{x}}^k (\Delta \bar{\mathbf{x}}^k)^T}{(\Delta \bar{\mathbf{x}}^k)^T \Delta \mathbf{G}^k} - \frac{\mathbf{A}^k \Delta \mathbf{G}^k (\Delta \mathbf{G}^k)^T (\mathbf{A}^k)^T}{(\Delta \mathbf{G}^k)^T (\mathbf{A}^k)^T \Delta \mathbf{G}^k}$$

This formula relates symmetry and positive definition of $\mathbf{A}^{k+1}$ in the absence of round-off-errors.

B) Constrained (equality) minimization cases

1) Gradient projection

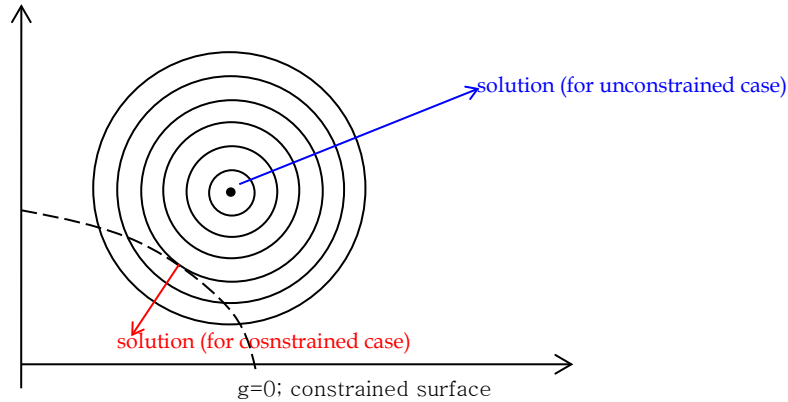
$$\begin{aligned}\text{Min } f(\bar{\mathbf{x}}), \quad \bar{\mathbf{x}} \in \mathbf{R}^N \\ \text{subject to } \bar{\mathbf{g}}(\bar{\mathbf{x}}) = 0, \quad \mathbf{g} \in \mathbf{R}^M, \quad M < N\end{aligned}$$

Necessary condition

$$\text{Lagrangian } L = f + \lambda^T \mathbf{g}$$

$$\frac{\partial L}{\partial \vec{x}} = 0 = \underbrace{\frac{\partial f}{\partial \vec{x}} + \frac{\partial \mathbf{g}^T}{\partial \vec{x}} \lambda}_{\text{projected gradient}}$$

$$\Delta \vec{x}^k = -\alpha \frac{\partial L^k}{\partial \vec{x}}$$



Starting with a feasible solution-

Step 1) Assume that we have a feasible solution

A feasible solution satisfied the constraints

$$\text{Define } \frac{\partial \mathbf{g}^T}{\partial \vec{x}} \equiv A, \quad A^T = \left[\frac{\partial g_i}{\partial \vec{x}_j} \right]_{M \times N \text{ matrix}}$$

$$\text{Solve } \frac{\partial L}{\partial \vec{x}} = \frac{\partial f}{\partial \vec{x}} + \frac{\partial \mathbf{g}^T}{\partial \vec{x}} \lambda = 0$$

$$A\lambda = -\frac{\partial f}{\partial \vec{x}}$$

$$\rightarrow \lambda = -\left(A^T A\right)^{-1} A^T \frac{\partial f}{\partial \vec{x}}$$

$$\text{Thus } \frac{\partial L}{\partial \vec{x}} = \left[I - A\left(A^T A\right)^{-1} A^T \right] \frac{\partial f}{\partial \vec{x}}$$

$$\Delta \vec{x}^k = -\alpha \frac{\partial L^k}{\partial \vec{x}} = -\alpha \left[I - A\left(A^T A\right)^{-1} A^T \right] \frac{\partial f}{\partial \vec{x}}$$

Step 2) Get back to the constraint ($g(\vec{x}^{k+1}) = 0$)

linearize the constraint

$$g(\vec{x}^k + \Delta\vec{x}^k) = g(\vec{x}^k) + A\Delta\vec{x}^k = 0$$

$$A\Delta\vec{x}^k = -g(\vec{x}^k) \quad \text{approximate solution}$$

$$\Delta\vec{x}^k = -\alpha (A^T A)^{-1} A^T g(\vec{x}^k)$$

Repeat Step 2) and 3) until convergence.

2) Quadratic Programming

$$\text{Min } f(\vec{x}) = \frac{1}{2} \vec{x}^T Q \vec{x} + c^T \vec{x}, \quad Q; \text{ symmetric positive definite}$$

$$\text{subject to } g(\vec{x}) = A\vec{x} - b = 0$$

(linear equality constraints)

Necessary condition

$$L(\vec{x}, \vec{\Lambda}) = f(\vec{x}) + \vec{\Lambda}^T (A\vec{x} - b)$$

$$\frac{\partial L}{\partial \vec{x}} = 0 = Q\vec{x} + c + \vec{\Lambda}^T A$$

$$\frac{\partial L}{\partial \vec{\Lambda}} = 0 = A\vec{x} - b$$

$$\begin{bmatrix} Q & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \vec{x} \\ \vec{\Lambda} \end{bmatrix} = \begin{bmatrix} -\vec{c} \\ b \end{bmatrix}$$

$$\begin{bmatrix} Q & A^T \\ A & 0 \end{bmatrix}^{-1} = \begin{bmatrix} H & P^T \\ P & S \end{bmatrix}$$

$$H = Q^{-1} - Q^{-1} A^T (A Q^{-1} A^T)^{-1} A Q^{-1}$$

$$P = (A Q^{-1} A^T)^{-1} A Q^{-1}$$

$$S = (A Q^{-1} A^T)^{-1}$$

$$\therefore \begin{bmatrix} \vec{x} \\ \vec{\Lambda} \end{bmatrix} = \begin{bmatrix} H & P^T \\ P & S \end{bmatrix} \begin{bmatrix} -\vec{c} \\ b \end{bmatrix}$$

$$\begin{aligned}\bar{\mathbf{x}} &= -\mathbf{H}\bar{\mathbf{c}} + \mathbf{P}^T \mathbf{b} \\ \bar{\boldsymbol{\Lambda}} &= -\mathbf{P}\mathbf{c} + \mathbf{S}\mathbf{b}\end{aligned}$$

3) Sequential Quadratic programming (SQP)

$$\text{Min } f(\bar{\mathbf{x}}) \quad \text{subject to } g_i(\bar{\mathbf{x}}) = 0, \quad i = 1, 2, \dots, m$$

$$f(\bar{\mathbf{x}}^k + \Delta \bar{\mathbf{x}}^k) \approx f(\bar{\mathbf{x}}^k) + \nabla f(\bar{\mathbf{x}}^k) \Delta \bar{\mathbf{x}}^k + \frac{1}{2} \Delta \bar{\mathbf{x}}^k H^k \Delta \bar{\mathbf{x}}^k$$

$$g_i(\bar{\mathbf{x}}^k + \Delta \bar{\mathbf{x}}^k) \approx g_i(\bar{\mathbf{x}}^k) + \nabla g_i(\bar{\mathbf{x}}^k) \Delta \bar{\mathbf{x}}^k$$

sub-problem (solved by Quadratic Programming)

$$\begin{aligned}\text{Min } & \nabla f(\bar{\mathbf{x}}^k) \Delta \bar{\mathbf{x}}^k + \frac{1}{2} \Delta \bar{\mathbf{x}}^k H^k \Delta \bar{\mathbf{x}}^k \\ & \text{subject to } g_i(\bar{\mathbf{x}}^k) + \nabla g_i(\bar{\mathbf{x}}^k) \Delta \bar{\mathbf{x}}^k = 0 \\ & i = 1, 2, \dots, m\end{aligned}$$

solve for $\Delta \bar{\mathbf{x}}^k$ and $\bar{\boldsymbol{\Lambda}}^k$ using Gradient Projection or Quadratic programming.